

HISQ gauge action conventions

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If I take the HISQ scaling paper at face value, then

$$S_{\text{MILC}}(U) = \frac{\beta_{\text{MILC}}}{3} \left[\sum_{n,\mu<\nu} \left(1 - \frac{1}{3} P_{\mu\nu} \right) - \frac{1 + 0.4805\alpha_s}{20u_0^2} \sum_{n,\mu\neq\nu} \left(1 - \frac{1}{3} R_{\mu\nu}(n) \right) + \mathcal{O}(\alpha_s) \right],$$

when $N_f = 0$

$$\beta_{\text{MILC}} = 10/g_0^2, \quad (1)$$

$$\alpha_s \propto \ln u_0. \quad (2)$$

As such,

$$S_{\text{MILC}}(U) = \beta_{\text{MILC}} \left[\sum_{n,\mu<\nu} \left(1 - \frac{1}{3} P_{\mu\nu} \right) - \frac{1}{20} \sum_{n,\mu\neq\nu} \left(1 - \frac{1}{3} R_{\mu\nu}(n) \right) \right] \quad (3)$$

when $u_0 = 1$. On the other hand, the **QEX** gauge action is normalized such that

$$S_{\text{QEX}}(U) = -\frac{\beta_{\text{QEX}}}{3} \left[c_p \sum_{n,\mu<\nu} P_{\mu\nu} + c_r \sum_{n,\mu\neq\nu} R_{\mu\nu}(n) \right] \quad (4)$$

yields the Lüscher-Weiss gauge action with

$$c_p + 8c_r = 1. \quad (5)$$

Note that QEX does not include the additional constant unit matrix that makes $S(U) = 0$ when $U = 1$. The classic Symanzik action follows when $c_r = -1/12$; that is,

$$S_{\text{LW}}(U) = -\frac{\beta_{\text{LW}}}{3} \left[\frac{5}{3} \sum_{n,\mu<\nu} P_{\mu\nu} - \frac{1}{12} \sum_{n,\mu\neq\nu} R_{\mu\nu}(n) \right]. \quad (6)$$

Assuming that MILC followed the same convention, I set

$$S_{\text{QEX}}(U) = -\frac{\beta_{\text{QEX}}}{3} \left[\sum_{n,\mu<\nu} P_{\mu\nu} - \frac{1}{20} \sum_{n,\mu\neq\nu} R_{\mu\nu}(n) \right] = S_{\text{MILC}} + C, \quad (7)$$

with C the constant that makes $S_{\text{MILC}}(U)$ vanish on unit configurations. which naively appears to violate Eqn. 5; however, that would only be true if the condition $\beta_{\text{QEX}} = 6/g_0^2$ were being imposed, **which it is not**. As such, one can safely conclude that

$$\beta_{\text{QEX}} = \beta_{\text{MILC}} \quad (8)$$

only because we have changed the definition of β_{QEX} by naively choosing $c_p = 1, c_r = -1/20$. It is still the case that Eqn. 5 is satisfied, as can be seen by rescaling:

$$S(U) = \frac{\beta}{5} \left[\frac{5}{3} \sum_{n,\mu<\nu} P_{\mu\nu} - \frac{1}{12} \sum_{n,\mu\neq\nu} R_{\mu\nu}(n) \right] = \frac{\beta}{5} \left[\tilde{c}_p \sum_{n,\mu<\nu} P_{\mu\nu} + \tilde{c}_r \sum_{n,\mu\neq\nu} R_{\mu\nu}(n) \right], \quad (9)$$

where

$$\tilde{c}_{\text{p}} = 5/3 = c_{\text{p}}, \tag{10}$$

$$\tilde{c}_{\text{r}} = -1/12 = c_{\text{r}}, \tag{11}$$

$$\beta \equiv \beta_{\text{QEX}} = \beta_{\text{MILC}} = 10/g_0^2. \tag{12}$$

It is otherwise the case that $\beta_{\text{QEX}} = 6/g_0^2$. **In other words, I got lucky with my guess.**